INCH-POUND
MIL-DTL-16377/81A
30 June 2014
SUPERSEDING
MIL-DTL-16377/81
18 December 2008

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; BALLAST, FOR ONE 17-WATT T8 FLUORESCENT LAMP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

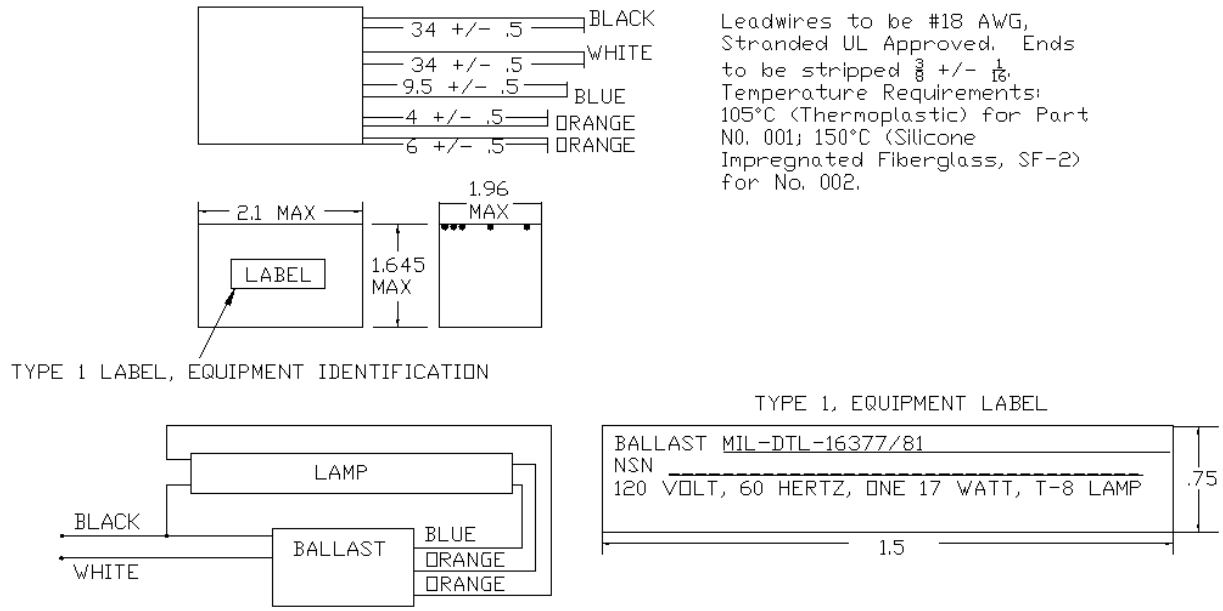
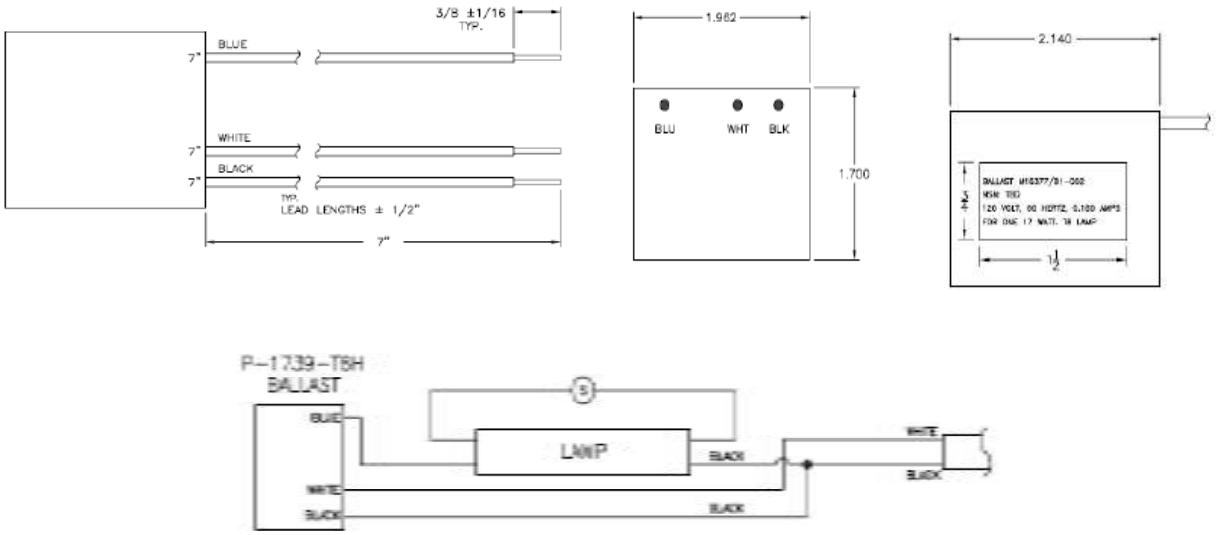


FIGURE 1. Ballast for one 17-watt T8 fluorescent lamp.

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NOTES:

1. Ballast lead wires to be #18 AWG, stranded UL approved, 302 °F (150 °C) silicone impregnated fiberglass insulation, cut and trimmed to dimensions shown.
2. Ballast shall use 40-watt FS-4 starter.
3. Rated for 248 °F (120 °C) environment.

FIGURE 2. High-temperature ballast for one 17-watt T8 fluorescent lamp.

REQUIREMENTS:

DESIGN:

Electrical rating:	17 watts, 120 volts, 60 hertz.
Power factor:	≥90 percent.
Efficiency:	>80 percent.
Circuit design M16377/81-001:	Ballast circuit shall incorporate intergraded electronic starter for lamp per-heat starter circuit.
Circuit design M16377/81-002:	Ballast circuit shall incorporate glow switch starter for lamp per-heat starter circuit.
Dimension and configurations:	See figures 1 and 2.
Impregnating:	Ballast circuitry shall be housed in thermoset plastic potting shell and potted with epoxy.
Designation and marking:	One Type I label, equipment identification, see MIL-DTL-16377 figure 11.

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EXAMINATIONS AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article testing/inspection:	See table II.
Conformance:	
Comparison inspection:	See table II.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION:

Table I. Standard parts.

Table II. First article tests.

TABLE I. Standard parts.

Name	Military P/N or Specification	Remarks
Ballast, 17-watt T8 fluorescent lamp, standard	M16377/81-001	Thermoplastic insulation
Ballast, 17-watt T8 fluorescent lamp, high-temperature	M16377/81-002	Silicone glass insulation

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TABLE II. First article tests and comparison inspection for ballast. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Electromagnetic interference	3.6.14	4.8.16	
Electrical equipment interface	3.6.18	4.8.21	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Shock	3.5.12	4.8.7	4/
Vibration	3.5.13	4.8.8	4/
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Noise	3.5.14	4.8.9	
Performance a. Power output b. Regulation c. Voltage drop across ballast d. Ballast efficiency e. Ballast power factor	5/	6/	7/

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ If less, then 1 KVA exception not applicable for input waveform total harmonic distortion.

3/ Solid state components shall be subject to 250 positive spikes and 250 negative spikes (500 total spikes), one spike per minute, using testing method identified in MIL-STD-1399-300.

4/ Tests shall be performed in the order specified.

5/ The performance of the ballast shall be in accordance with the requirements listed in the applicable paragraph of NEMA ANSI C82.2.

6/ Test procedure for performance shall be as follows:

a. Performance tests shall be in accordance with NEMA ANSI C82.2.

b. Fluorescent lamp reference ballasts shall be in accordance with NEMA ANSI C82.3.

c. Lamps shall conform to ANSI/ANSLG C78.81.

d. Voltage across starter terminals and across ballast, when supplying a reference lamp, shall be measured.

7/ The sample lot size for inspection shall consist of five ballast units.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR4
Navy – SH
Air Force – 99

Preparing activity:
Navy – SH
(Project 6250-2012-002)

Review activities:
Navy – AS, CG
Air Force – 03
DLA – GS2

Civil agency:
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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